Bias reduction for local linear estimator

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Nov 14, 2025

model

Consider the nonparametric regression model

$$Y = m(X) + \sigma(X)e$$

where $m(x) = E(Y|X = x), \sigma^2(x) = Var(Y|X = x)$, e is a random error independet of X with mean zero and variance 1.
Suppose the data $(X_i, Y_i), i = 1, ..., n$ are observed from the model.

Local linear estimator

Let $\hat{m}_h(x)$ be a local linear estimator that is obtained by minimizing

$$\sum_{i=1}^{n}\{Y_i - a - b(X_i - x)\}^2 K_h(X_i - x)$$

If $x \in [h, 1-h]$, $m^{(2)}, f, \sigma^2$ is continuous at x, $f(x) > 0$, K is symmetric,

$$\mathbb{E}(\hat{m}_h(x)|\mathbb{X}) - m(x) = \frac{1}{2}\mu_2 m^{(2)}(x)h^2 + o_p(h^2)$$

$$Var(\hat{m}_h(x)|\mathbb{X}) = \nu_0 \frac{\sigma^2(x)}{f(x)nh} + o_p(\frac{1}{nh})$$

where $\mu_2 = \int u^2 K(u)du,\ \nu_0 = \int K^2(u)du$

Reducing bias

Motivated by

$$\mathbb{E}(\hat{m}_h(x)|\mathbb{X}) - m(x) = \frac{1}{2}\mu_2 m^{(2)}(x)h^2 + o_p(h^2),$$

consider a linear regression model

$$\hat{m}_h(x) \approx \alpha + \beta h^2 + \sqrt{Var(\hat{m}_h(x))}\varepsilon_i$$

to estimate bias reduced estimator, where ε is independent of h_i with $E\varepsilon_i = 0, Var\varepsilon_i = 1$.

Then we can estimate bias reduced $\hat{m}(x)$ by $\hat{\alpha}$, and $\hat{\beta}$ estimates $\frac{1}{2}\mu_2 m^{(2)}(x)$.

Reducing bias

Given bandwidths $h_1, ..., h_B$, we denote respective local linear estimators by

$$V_1 \equiv \hat{m}_{h_1}(x), ..., V_B \equiv \hat{m}_{h_B}(x)$$

Consider linear regression model for $(V_i, h_i), i = 1, ..., B$,

$$V_i \approx \alpha + \beta h_i^2 + \tilde{\sigma}(h_i)\varepsilon_i$$

the least squares estimators are

$$\hat{\alpha}_B = \sum_{i=1}^{B} \left(\frac{\sum_{k=1}^{B} h_k^4 - (\sum_{k=1}^{B} h_k^2) \cdot h_i^2}{B \sum_{k=1}^{B} h_k^4 - (\sum_{k=1}^{B} h_k^2)^2} \right) V_i, \quad \hat{\beta}_B = \sum_{i=1}^{B} \left(\frac{B \cdot h_i^2 - (\sum_{k=1}^{B} h_k^2)}{B \sum_{k=1}^{B} h_k^4 - (\sum_{k=1}^{B} h_k^2)^2} \right) V_i$$

Reducing bias

Now we denote the bias reduced estimator by $\tilde{m}_h(x)$.

$$\tilde{m}_B(x) = \hat{\alpha}_B(x) = \sum_{i=1}^{n} \mathbf{g}_i V_i$$

where

$$\mathbf{g}_i = \frac{\sum_{k=1}^{B} h_k^4 - (\sum_{k=1}^{B} h_k^2) \cdot h_i^2}{B \sum_{k=1}^{B} h_k^4 - (\sum_{k=1}^{B} h_k^2)^2}, i = 1, ..., B.$$

Assumptions

(a) For an $s > 2$, $E|Y|^{2s} < \infty$ and $E|X|^{2s} < \infty$.
(b) The density function of $X, f(x)$, is continuous and positive on its compact support.
(c) The second derivatives of $f(x)$ and $\sigma^2(x)$ are continuous and bounded.
(d) The fourth derivative of $m(x), m^{(4)}(x)$, is continuous.
(e) The kernel function $K(t)$ is a symmetric density function and is absolutely continuous on its support set $[-A, A]$.

Lemma 1

Under assumptions (a)-(e), we have the following higher-order expansion of bias of local linear estimator.

$$E(\tilde{m}_h(x_0)|\mathbf{X}) - m(x_0) = \frac{1}{2}\mu_2 m^{(2)}(x_0)h^2 + \mathbf{a}(x_0)h^4 + o_p(h^4)$$

where

$$\mathbf{a}(x_0) = \frac{1}{24}\mu_4 m^{(4)}(x_0) - \frac{m^{(2)}(x_0)}{2}\mathbf{b}(x_0)\mu_4,$$
$$\mathbf{b}(x_0) = \left(\frac{f^{(1)}(x_0)}{f(x_0)}\right)^2.$$

Proof of Lemma 1

$$E[\tilde{m}_h(x_0)|\mathbf{X}] - m(x_0) = e_1^T(\frac{1}{n}\mathbf{D}_{x_0}^T\mathbf{W}_{x_0}\mathbf{D}_{x_0})^{-1}\frac{1}{n}\mathbf{D}_{x_0}^T\mathbf{W}_{x_0}(\mathbf{Y} - \mathbf{D}_{x_0}m(x_0))$$

$$\begin{aligned}
\frac{1}{n}\mathbf{D}_{x_0}^T\mathbf{W}_{x_0}\mathbf{D}_{x_0} &= \begin{bmatrix} \frac{1}{n}\sum K_h(X_i - x_0) & \frac{1}{n}\sum K_h(X_i - x_0)\cdot(X_i - x_0) \\ \frac{1}{n}\sum K_h(X_i - x_0)\cdot(X_i - x_0) & \frac{1}{n}\sum K_h(X_i - x_0)\cdot(X_i - x_0)^2 \end{bmatrix} \\
&= \begin{bmatrix} \mu_0 f(x_0) + \mu_1 f(x_0)h & \mu_1 f(x_0) + \mu_2 f'(x_0)h \\ \mu_1 f(x_0) + \mu_2 f'(x_0)h & \mu_2 f(x_0) + \mu_3 f'(x_0)h \end{bmatrix} + o_p(h) \\
&= f(x_0)\begin{bmatrix} 1 & 0 \\ 0 & \mu_2 \end{bmatrix} + hf'(x_0)\begin{bmatrix} 0 & \mu_2 \\ \mu_2 & 0 \end{bmatrix} + o_p(h)
\end{aligned}$$

Proof of Lemma 1

Using

$$(A + hB + o_p(h))^{-1} = (I + hA^{-1}B + A^{-1}o_p(h))^{-1}A^{-1}$$
$$= A^{-1} - hA^{-1}BA^{-1} + o_p(h)$$

$$\frac{1}{n}\mathbf{D}_{x_0}^T\mathbf{W}_{x_0}\mathbf{D}_{x_0} = \frac{1}{f(x_0)}\begin{bmatrix}1 & 0\\ 0 & 1/\mu_2\end{bmatrix} - \frac{hf'(x_0)}{f(x_0)^2}\begin{bmatrix}1 & 0\\ 0 & 1/\mu_2\end{bmatrix}\begin{bmatrix}0 & \mu_2\\ \mu_2 & 0\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1/\mu_2\end{bmatrix} + o_p(h)$$

Proof of Lemma 1

$$
\begin{aligned}
&\frac{1}{n}\mathbf{D}_{x_0}^T\mathbf{W}_{x_0}(\mathbf{Y}-\mathbf{D}_{x_0}m(x_0)) \\
&=\frac{1}{n}\mathbf{D}_{x_0}^T\mathbf{W}_{x_0}\left(\frac{m^{(2)}(x_0)}{2!}\begin{bmatrix}(X_1-x_0)^2\\ \vdots \\ (X_n-x_0)^2\end{bmatrix}+\cdots\frac{m^{(4)}(x_0)}{4!}\begin{bmatrix}(X_1-x_0)^4\\ \vdots \\ (X_n-x_0)^4\end{bmatrix}\right)
\end{aligned}
$$

Bias and variance of $\tilde{m}_B(x)$

Theorem 1. Under assumptions (a)-(e), $h_i = C_i h, i = 1, ..., B$ with $h \to 0$ and $nh \to \infty$ as $n \to \infty$, we have

$$E\{\tilde{m}_B(x_0)|\mathbf{X}\} = m(x_0) + \mathbf{C}(x_0)h^4 + o_p(h^4)$$

$$Var\{\tilde{m}_B(x_0)|\mathbf{X}\} = \frac{\sigma^2(x_0)}{nf(x_0)} \sum_{i=1}^{B} \sum_{j=1}^{B} \mathbf{g}_i \mathbf{g}_j \{\psi_{ij}^{(0)} + o_p(\frac{1}{h})\}$$

where

$$\mathbf{C}(x_0) = \frac{1}{2}\mathbf{d}(x_0)\left(\frac{1}{12}m^{(4)}(x_0) - m^{(2)}(x_0)\mathbf{b}(x_0)\right)\mu_4 = \mathbf{a}(x_0)\mathbf{d}(x_0)$$

$$\mathbf{d}(x_0) = \frac{\sum_{k=1}^{B} C_k^4 \sum_{i=1}^{B} C_i^4 - \sum_{k=1}^{B} C_k^2 \sum_{i=1}^{B} C_i^6}{B\sum_{k=1}^{B} C_k^4 - (\sum_{k=1}^{B} C_k^2)^2}$$

$$\psi_{ij}^{(k)} = \int K(h_i u)K(h_j u)u^k du.$$

proof of bias of $\tilde{m}_B(x)$

Using the result of lemma 1,

$$
\begin{aligned}
&E[\hat{\alpha}_B(x_0) - m(x_0)|\mathbf{X}] \\
&= \sum_{i=1}^{B} \mathbf{g}_i \times [\frac{1}{2}\mu_2 m^{(2)}(x_0)h_i^2 + \mathbf{a}(x_0)h_i^4] + o_p(h^4) \\
&= \frac{1}{2}\mu_2 m^{(2)}(x_0) \times \sum_{i=1}^{B} \mathbf{g}_i h_i^2 + \mathbf{a}(x_0) \times \sum_{i=1}^{B} \mathbf{g}_i h_i^4 + o_p(h^4) \\
&= 0 + \mathbf{C}(x_0)h^4 + o_p(h^4)
\end{aligned}
$$

proof of variance of $\tilde{m}_B(x)$

We have

$$Var(\hat{\alpha}_B(x_0)|\mathbf{X}) = \sum_{i=1}^{B} \mathbf{g}_i^2 \left(\nu_0 \frac{\sigma^2(x_0)}{f(x_0)nh_i} + o_p(\frac{1}{nh}) \right) + 2\sum_{i<j}^{B} \mathbf{g}_i \mathbf{g}_j Cov(V_i, V_j|\mathbf{X})$$

We need to calculate $Cov(V_i, V_j|\mathbf{X})$.

$$\begin{aligned} V_i - EV_i &= e_1^T(\frac{1}{n}D_x^T W_x D_x)^{-1}\frac{1}{n}D_x W_x(\mathbf{Y} - \mathbf{m}) \\ &= e_1^T \left(f(x_0)\mathbf{M} + \mathbf{o_p(1)}\right)^{-1} \frac{1}{n}D_x W_x(\mathbf{Y} - \mathbf{m}) \\ &= \frac{1}{nf(x_0)} \sum_{l=1}^{n} K_{h_i}(X_l - x_0)(Y_l - m(X_l)) \cdot (1 + o_p(1)) \end{aligned}$$

proof of variance of $\tilde{m}_B(x)$

Therefore,

$$
\begin{aligned}
Cov(V_i, V_j|\mathbf{X}) &= E[(V_i - EV_i)(V_j - EV_j)|\mathbf{X}] \\
&= \frac{\sigma^2(x_0)}{n^2 f(x_0)^2} \sum_{l=1}^{n} K_{h_i}(X_l - x_0) K_{h_j}(X_l - x_0) \cdot (1 + o_p(1)) \\
&= \frac{\sigma^2(x_0)}{n f(x_0)^2} \big(E K_{h_i}(X - x_0) K_{h_j}(X - x_0)\big) \cdot (1 + o_p(1)) \\
&= \frac{\sigma^2(x_0)}{n f(x_0)^2} \int K(h_i u) K(h_j u) f(x_0) du \cdot (1 + o_p(1)) \\
&= \frac{\sigma^2(x_0)}{n f(x_0)} (\psi_{ij}^{(0)} + o_p(1))
\end{aligned}
$$

proof of variance of $\tilde{m}_B(x)$

Using

$$\psi_{ii}^{(0)} = \frac{1}{h_i}\int K^2(u)du = \frac{\nu_0}{h_i},$$

We have

$$\begin{aligned} Var(\hat{\alpha}_B(x_0)|\mathbf{X}) &= \sum_{i=1}^{B} \mathbf{g}_i^2 \left(\nu_0 \frac{\sigma^2(x_0)}{f(x_0)nh_i} + o_p(\frac{1}{nh})\right) + 2\sum_{i<j}^{B} \mathbf{g}_i\mathbf{g}_j Cov(V_i, V_j|\mathbf{X}) \\ &= \frac{\sigma^2(x_0)}{nf(x_0)} \sum_{i=1}^{B}\sum_{j=1}^{B} \mathbf{g}_i\mathbf{g}_j(\psi_{ij}^{(0)} + o_p(\frac{1}{h})) \end{aligned}$$

Remark

- We can use this method when $x_0 \in [\max\{h_1, ..., h_B\}, 1 - \max\{h_1, ..., h_B\}]$
  If $x_0 \in [0, \max\{h_1, ..., h_B\}]$,

$$\begin{aligned} E[\tilde{m}_h(x_0)|\mathbf{X}] - m(x_0) &= e_1^T h^2 \frac{1}{2} \mathbf{M}(x) \boldsymbol{\mu}(x) m^{(2)}(x_0) + o_p(h^2) \\ &= \frac{1}{2} \frac{\mu_2(x_0)^2 - \mu_1(x_0)\mu_3(x_0)}{\mu_2(x_0)\mu_0(x_0) - \mu_1(x_0)^2} h^2 + o_p(h^2) \end{aligned}$$

  We need to consider following regression model for bias reduction

$$V_i = \alpha + \beta \frac{\mu_{2,c_i}^2 - \mu_{1,c_i}\mu_{3,c_i}}{\mu_{2,c_i}\mu_{0,c_i} - \mu_{1,c_i}^2} h_i^2 + \varepsilon_i$$

  where $\mu_{j,c_i} = \int_{-c_i}^{\infty} u^j K(u) du$ and $c_i = x_0 / h_i$.

Another bias reduced estimator

Since $\hat{\beta}_B$ estimates $(1/2)\mu_2 m^{(2)}(x_0)$, we can consider other bias reduced estimator

$$\hat{m}_B(x_0) = \hat{m}_h(x_0) - \hat{\beta}_B h^2$$

Asymptotic bias and variance are as following

$$E(\hat{m}_B(x_0)|\mathbf{X}) - m(x_0) = O_p(h^4 + h^2 h_0^2)$$

$$Var(\hat{m}_B(x_0)|\mathbf{X}) = Var(\hat{m}_h(x_0)) + O_p\left(\frac{h^4}{nh_0^5} + \frac{h^2}{n\sqrt{hh_0^5}}\right)$$

where $h_i = C_i h_0$.

Another bias reduced estimator

Note that

$$
\begin{aligned}
E(\hat{\beta}_B|\mathbf{X}) &= \sum_{i=1}^{B}\left(\frac{B\cdot h_i^2-(\sum_{k=1}^{B}h_k^2)}{B\sum_{k=1}^{B}h_k^4-(\sum_{k=1}^{B}h_k^2)^2}\right)\left(m(x_0)+\frac{1}{2}\mu_2 m^{(2)}(x_0)h^2+\mathbf{a}(x_0)h^4+o_p(h^4)\right)\\
&= \frac{1}{2}\mu_2 m^{(2)}(x_0)+O_p(h^4)
\end{aligned}
$$

Let $h_i = C_i h_0$. Then

$$
\begin{aligned}
&E(\hat{m}_B(x_0)|\mathbf{X})-m(x_0)\\
&= E(\hat{m}_h(x_0)|\mathbf{X})-m(x_0)-E(\hat{\beta}_B|\mathbf{X})h^2\\
&= \frac{1}{2}\mu_2 m^{(2)}(x_0)h^2+O_p(h^4)-h^2(\frac{1}{2}\mu_2 m^{(2)}(x_0)+O_p(h_0^2))\\
&= O_p(h^4+h^2h_0^2)
\end{aligned}
$$

Another bias reduced estimator

$$Var(\hat{m}_B(x_0)|\mathbf{X}) = Var(\hat{m}_h(x_0)|\mathbf{X}) + Var(\hat{\beta}_B|\mathbf{X})h^4 - 2Cov(\hat{m}_h(x_0), \hat{\beta}_B|\mathbf{X})h^2$$

Through some calculation,

$$Var(\hat{\beta}_B|\mathbf{X}) = O_p\left(\frac{1}{h_0^4}\right) O_p\left(\frac{1}{nh_0}\right) = O_p\left(\frac{1}{nh_0^5}\right)$$

$$\begin{aligned} Cov(\hat{m}_h(x_0), \hat{\beta}_B|\mathbf{X}) &= O_p\left(\frac{1}{h_0^2}\right) \frac{1}{n} \int K(hu)K(h_0u)du \\ &\leq O_p\left(\frac{1}{h_0^2}\right) \frac{1}{n} \sqrt{\int K^2(hu)du \int K^2(h_0u)du} \\ &= O_p\left(\frac{1}{n\sqrt{hh_0^5}}\right) \end{aligned}$$

Another bias reduced estimator

If we take h such that $h = o(h_0)$,

$$\begin{aligned}
Var(\hat{m}_B(x_0)|\mathbf{X}) &= Var(\hat{m}_h(x_0)|\mathbf{X}) + Var(\hat{\beta}_B|\mathbf{X})h^4 - 2Cov(\hat{m}_h(x_0), \hat{\beta}_B|\mathbf{X})h^2 \\
&= \nu_0 \frac{\sigma^2(x_0)}{nhf(x_0)} + o_p\left(\frac{1}{nh}\right) + O_p\left(\frac{h^4}{nh_0^5} + \frac{h^2}{n\sqrt{hh_0^5}}\right) \\
&= \nu_0 \frac{\sigma^2(x_0)}{nhf(x_0)} + o_p\left(\frac{1}{nh}\right)
\end{aligned}$$

Application to SBF

Using N-W estimator

$$\hat{m}_j(x_j) = m_j(x_j) + \tilde{m}_j^A(x_j) + \frac{1}{2}h_j^2\mu_2 m''(x_j) + \Delta_j(x_j) + o_p(n^{-2/5})$$

Using local linear

$$\hat{m}_j^{tp}(x_j) = m_j^{tp}(x_j) + \tilde{m}_j^{A,tp}(x_j) + \frac{1}{2}h_j^2\mathbf{N}_j(x_j)^{-1}\boldsymbol{\gamma}_j(x_j)m''(x_j) + o_p(n^{-2/5})$$

where $(\tilde{\mathbf{m}}_j^{A,tp}(x_j))_l = \hat{\mathbf{M}}_{jj}(x_j)^{-1}\frac{1}{n}\sum_{i=1}^n\left(\frac{X_{ij}-x_j}{h_j}\right)^l K_{h_j}(x_j, X_{ij})\cdot\epsilon_i$
and $(\tilde{\mathbf{m}}_j^{A,tp}(x_j))_l = O_p(\frac{1}{\sqrt{nh_j}}) = O_p(h_j^2), 0 \le l \le r$.